

Acute Interstitial Nephritis

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DEFINITION

Acute interstitial nephritis (AIN) is an acute, often reversible disease characterized by inflammatory infiltrates within the kidney interstitium. AIN is a relatively rare cause of acute kidney injury (AKI), but it should not be overlooked because it usually requires specific therapeutic interventions.

PATHOGENESIS

Most studies suggest that AIN is an immunologically induced hypersensitivity reaction to an antigen that is classically a drug or an infectious agent. Evidence for a hypersensitivity reaction in drug-induced AIN includes the following: it occurs only in a small percentage of individuals; it is not dose dependent; it is often associated with extrarenal manifestations of hypersensitivity; it recurs after accidental reexposure to the same drug or to a closely related one; and it is sometimes associated with evidence of delayed-type hypersensitivity reaction (kidney granulomas). Similarly, AIN secondary to infections can be differentiated from pyelonephritis by the relative absence of neutrophils in the interstitial infiltrates and the failure to isolate the infective agent from the kidney parenchyma, again suggesting an immunologic basis to the disease.

Experimental models of AIN have identified three major types of antigens that induce the condition.¹ Antigens may be tubular basement membrane (TBM) components (e.g., the glycoproteins 3M-1 and TIN-Ag/TIN1), secreted tubular proteins (e.g., uromodulin), or nonrenal proteins (e.g., from immune complexes).

Although human AIN may be secondary to an immune reaction directed against a kidney antigen, most cases of AIN are probably induced by extrarenal antigens, being produced in particular by drugs or infectious agents. These antigens may induce AIN by a variety of mechanisms, including binding to kidney structures ("planted antigen"); acting as haptens that modify the immunogenicity of native kidney proteins; mimicking kidney antigens, resulting in a cross-reactive immune reaction; and precipitating within the interstitium as circulating immune complexes.

Experimental models have identified both cell-mediated immunity and antibody-mediated immunity in the pathogenesis of AIN (Fig. 64.1). In humans, most forms of AIN are not associated with antibody deposition, which suggests that cell-mediated immunity plays a major role. This hypothesis is reinforced by the fact that interstitial infiltrates

usually contain numerous T cells, and these infiltrates sometimes form granulomas. Nevertheless, deposition of anti-TBM antibodies or of immune complexes can be observed occasionally in kidney biopsy specimens, and antibody-mediated immunity may play a role in the pathogenesis of the disease in these cases.

Formation of immune complexes within the interstitium, or interstitial infiltration with T cells, will result in an inflammatory reaction. This reaction is triggered by many events, including activation of the complement cascade by antibodies and release of inflammatory cytokines by T lymphocytes and phagocytes (see Fig. 64.1). Although the interstitial inflammatory reaction may resolve without sequelae, it sometimes induces interstitial fibroblast proliferation and extracellular matrix synthesis, leading to interstitial fibrosis and chronic kidney disease (CKD). Cytokines such as transforming growth factor- β appear to play a key role in the latter process.

EPIDEMIOLOGY

AIN is a relatively uncommon cause of AKI, although its incidence may be increasing.² It accounts for 2% to 3% of all kidney biopsies, 10% to 15% of biopsies performed for unexplained AKI, and up to 25% of those performed for drug-induced AKI.² Of kidney biopsies done in children with AKI, 3% to 7% show AIN.³

Before antibiotics were available, AIN was most commonly associated with infections such as scarlet fever and diphtheria. Drug-induced AIN now appears to account for about 70% to 90% of all cases.

DRUG-INDUCED ACUTE INTERSTITIAL NEPHRITIS

Clinical Manifestations

In the 1960s and 1970s, most cases of drug-induced AIN were caused by methicillin, and the clinical manifestations of methicillin-induced AIN were considered the prototypical presentation of AIN. Since then, many other drugs have been implicated in the induction of AIN (Box 64.1), of which antimicrobial agents (in particular, β -lactam antibiotics, sulfonamides, fluoroquinolones, and rifampin), nonsteroidal antiinflammatory drugs (NSAIDs) (especially fenoprofen), and proton pump inhibitors (PPIs) have been most commonly involved.^{4,5} In a large case series of biopsy-proven AIN, drug-induced AIN was because of antibiotics in 49% of cases, NSAIDs in 11%, and PPIs in 14%.⁴ There has been a rising incidence of AIN in the current decade. In a large single-center Spanish

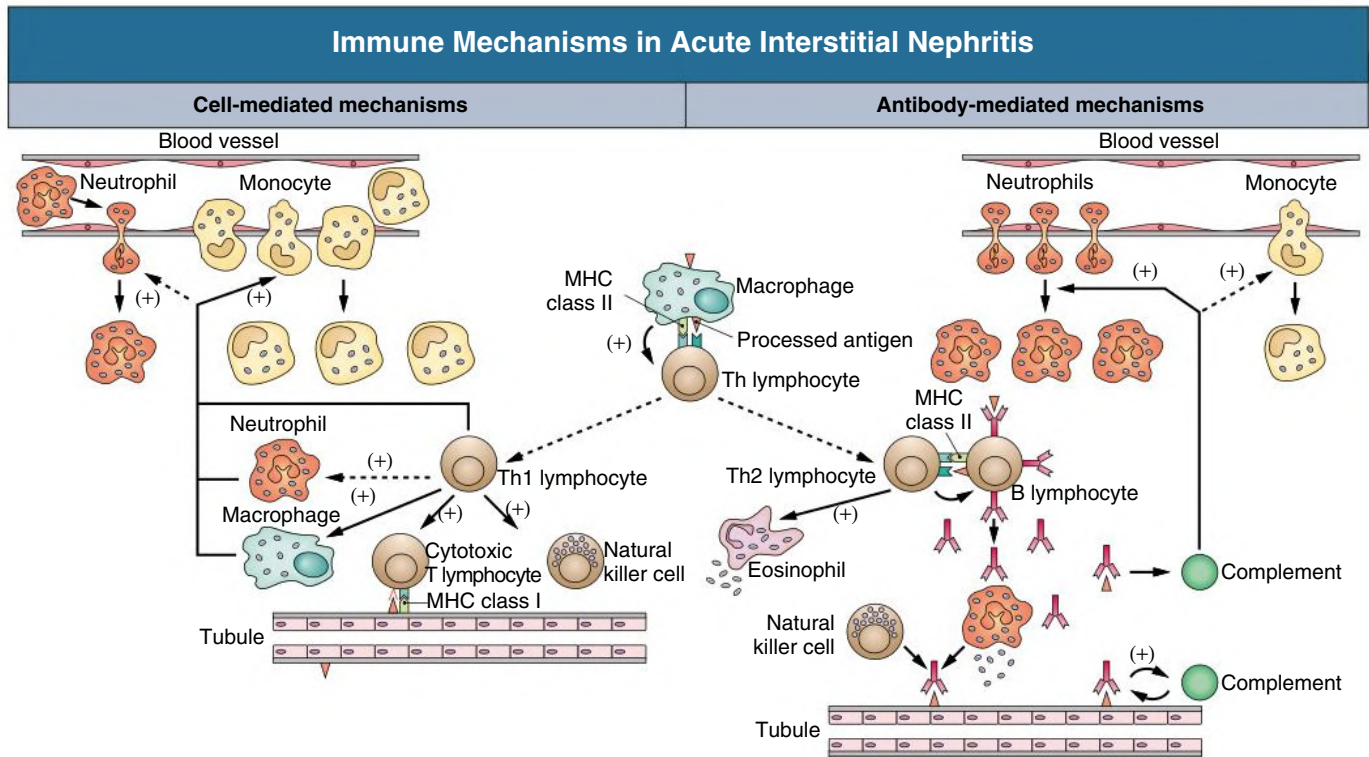


Fig. 64.1 Immune Mechanisms That Can Be Involved in Acute Interstitial Nephritis. Both cell-mediated and antibody-mediated mechanisms occur. The cell-mediated mechanism is primarily associated with macrophages and T cells. The antibody-mediated mechanism is frequently associated with neutrophil or eosinophil infiltration and with local complement activation. *MHC*, Major histocompatibility complex.

study of 182 patients, the most common offending drugs were NSAIDs (27%), antibiotics (22%), and PPIs (4%), but 30% of cases had no identified offending drug.⁶ Other antiulcer agents, diuretics, phenindione, phenytoin, allopurinol, highly active antiretroviral therapy (HAART), and anticancer agents such as tyrosine kinase inhibitors, pemetrexed, and ifosfamide also have been reported to cause AIN.^{7,8} In addition, AIN can be caused by immune checkpoint inhibitors (e.g., programmed death 1 [PD-1], PD ligand 1 [PD-L1], and cytotoxic T-lymphocyte-associated antigen 4 [CTLA-4] inhibitors), as shown in meta-analysis and now confirmed by a large single-center study of 1016 patients to occur about 3% of the time.^{9,10} Most other drugs only rarely have been linked with AIN (see Box 64.1). AIN also has been reported after snake or wasp envenomation, and the pathogenic mechanism is likely to be similar to that in drug-induced AIN.

The clinical characteristics of drug-induced AIN are now recognized as much more varied and nonspecific than the spectrum seen in classic methicillin-induced AIN (Fig. 64.2).^{2,11,12}

Kidney Manifestations

Symptoms of AIN usually develop a few days to a few weeks after the inciting drug is started, although cases have occurred months after initial exposure to the drug. In particular, AIN induced by NSAIDs or PPIs is often diagnosed several months after treatment initiation, and a recent paper suggests an average time of 4 months with immune checkpoint inhibitors.¹¹ The typical presentation is sudden impairment in kidney function, associated with mild proteinuria (<1 g/day)

and abnormal urinalysis, in a patient with flank pain, normal blood pressure, and no edema. Clinical presentation is often incomplete (see Fig. 64.2), and AIN should be considered in any patient with unexplained AKI.² Kidney dysfunction may be mild or severe, and dialysis is required in about one-third of patients. Hematuria and pyuria are present in a little more than half of the patients, and although leukocyte casts are common, hematuria is almost never associated with red blood cell casts. Flank pain (reflecting distention of the kidney capsule) is observed in about one-third of the patients and can be the main complaint on hospital admission. Some patients have low fractional sodium excretion.

Standard imaging procedures show kidneys normal in size or slightly enlarged. Ultrasound usually discloses an increased cortical echogenicity (comparable to or higher than that of the liver).

Extrarenal Manifestations

Extrarenal symptoms consistent with a hypersensitivity reaction are occasionally observed, including low-grade fever, maculopapular rash (Fig. 64.3), mild arthralgia, and eosinophilia. However, each of these symptoms is typically present in fewer than half of the patients (see Fig. 64.2), and these symptoms are all present in less than 10% of patients.² With some drugs, other manifestations of hypersensitivity, such as hemolysis or hepatitis, can be present. Serum immunoglobulin E (IgE) levels also may be elevated.

The association of AKI with clinical signs suggestive of hypersensitivity or eosinophilia should lead to consideration of AIN. However, signs of hypersensitivity can be observed in patients with AKI that is

BOX 64.1 Drugs Responsible for Acute Interstitial Nephritis

Antimicrobial Agents**Penicillins**

Amoxicillin

Ampicillin^a

Aztreonam

Carbenicillin

Cloxacillin

Flucloxacillin

Methicillin^a

Mezlocillin

Nafcillin

Oxacillin^a**Benzylpenicillin^a**

Piperacillin

Cephalosporins

Cefaclor

Cefazolin

Cefotaxime

Cefotetan

Cefoxitin

Ceftriaxone

Cephalexin

Cephaloridine

Cephalothin

Cephadrine

Quinolones**Ciprofloxacin^a**Levofloxacin^a

Moxifloxacin

Norfloxacin

Other

Acyclovir

Azithromycin

Clarithromycin

Colistin

Cotrimoxazole^aErythromycin^a

Ethambutol

Flurithromycin

Foscarnet

Gentamicin

Indinavir

Interferon

Isoniazid

Lincomycin

Linezolid

Minocycline

Nitrofurantoin^a

Piromidic acid

Polymyxin B^a

Quinine

Rifampin^a (rifampicin^a)Spiramycin^a**Sulfonamides^a**

Teicoplanin

Telithromycin

Tetracycline

Vancomycin^a**NSAIDs, Including Salicylates****Salicylates and Derivatives****Acetylsalicylic acid (aspirin)**Diflunisal^a**Propionic Acid Derivatives**

Benoxaprofen

Fenbufen

Fenoprofen^a

Flurbiprofen

Ibuprofen^a

Ketoprofen

Naproxen**Acetic Acid Derivatives****Indometacin^a (indomethacin^a)**

Alclofenac

Diclofenac

Fenclofenac

Sulindac

Zomepirac**Enolic Acid Derivatives**

Meloxicam

Piroxicam^a**Fenamic Acid Derivatives****Mefenamic acid****Coxibs**

Celecoxib

Rofecoxib

Other

Azapropazone

Mesalamine (mesalazine, 5-ASA)

Phenazone

Phenylbutazone

Sulfasalazine

Tolmetin**Antalgics**

Aminopyrine

Antipyrine

AnticonvulsantsCarbamazepine^a

Diazepam

Lamotrigine^a

Levetiracetam

Phenobarbital (phenobarbitone)

Phenytoin^a

Valproic acid (valproate sodium)

Diuretics

Chlorthalidone

Etacrynic acid (ethacrynic acid)

Furosemide^a (frusemide^a)Hydrochlorothiazide^a

Indapamide

Triamterene^a**Antiulcer Agents****Histamine-2 Receptor****Antagonists****Cimetidine^a**

Famotidine

Ranitidine

Proton Pump Inhibitors

Esomeprazole

Lansoprazole

Omeprazole

Pantoprazole

Rabeprazole

Immune Checkpoint**Inhibitors**Ipilimumab^a

Nivolumab

Pembrolizumab

Other Anticancer Agents

Adriamycin

Bevacizumab

Bortezomib

Carboplatin

Cytosine arabinoside

Gemcitabine

Ifosfamide

Interleukin-2

Lenalidomide

Methotrexate

Pemetrexed

Sorafenib

Sunitinib

Others**Allopurinol^a**

α-Methyl dopa

Amlodipine

Azathioprine

Betanidine (bethanidine)^a

Bismuth salts

Captopril^a

Carbimazole

Cetirizine

Chlorpropamide^a

Clofibrate

Clozapine

Cyclosporine

Deferasirox

Diltiazem

D-Penicillamine

Etanercept

Exenatide

Fenofibrate^a

Fluindione

Gold salts

Griseofulvin

Interferon

Isotretinoin

Liraglutide

Nifedipine^a**Phenindione^a**

Phenothiazine

Phenylpropanolamine

Probenecid

Propranolol

Propylthiouracil

Sirolimus

Streptokinase

Sulfinpyrazone

Warfarin

Zopiclone

Drugs most commonly involved are in **bold**.^aDrugs that can cause granulomatous AIN.

AIN, Acute interstitial nephritis; NSAIDs, nonsteroidal antiinflammatory drugs.

not related to AIN, including patients with drug-induced acute tubular necrosis.

Specific Drug Associations

The clinical and biologic manifestations of AIN may have some specificity, depending on the drug involved.

As outlined earlier, methicillin-induced AIN was characterized by a high frequency of abnormal urinalysis and extrarenal symptoms and by well-preserved kidney function. Kidney failure was reported in only about 50% of patients (see Fig. 64.2).²

More than 200 cases of rifampin-induced AKI have been reported.¹² Prior exposure is common; most cases occurred after readministration

Clinical Manifestations of Drug-Induced Acute Interstitial Nephritis

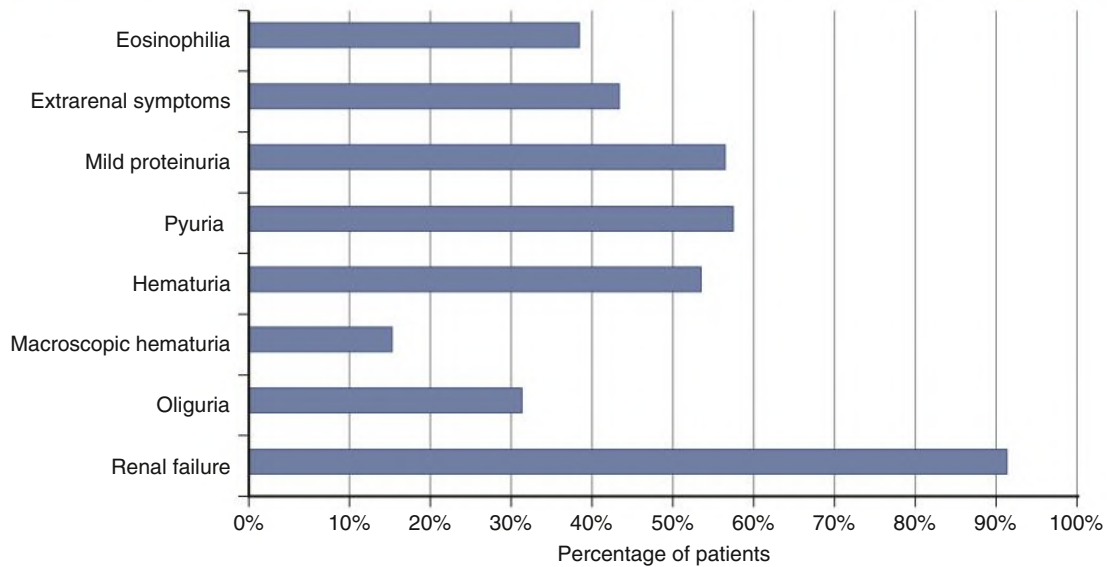


Fig. 64.2 Clinical Manifestations of Drug-Induced Acute Interstitial Nephritis (AIN). Data were pooled from different case reports, including more than 200 patients with drug-induced AIN. Patients with AIN associated with methicillin therapy or with a nephrotic syndrome are not included.



Fig. 64.3 Maculopapular Rash in a Patient With Drug-Induced Acute Interstitial Nephritis (AIN). Such cutaneous lesions occur in about 40% of patients with drug-induced AIN, but they can also be seen in patients with drug-induced acute tubular necrosis.

of rifampin or several months after intermittent administration of the drug had begun. Kidney failure is usually associated with the sudden onset of fever, gastrointestinal symptoms (nausea, vomiting, diarrhea, abdominal pain), and myalgias. It also may be associated with

hemolysis, thrombocytopenia, and, less frequently, hepatitis. Kidney biopsy usually shows tubular injury and interstitial inflammatory infiltrates. Although circulating antirifampin antibodies are often found, immunofluorescence (IF) staining of kidney biopsy specimens for immunoglobulin and complement is negative in most cases, suggesting that cell-mediated immunity plays a key role in the induction of the nephritis. In a few cases, AIN developed after continuous treatment with rifampin for 1 to 10 weeks in which extrarenal symptoms or antirifampin antibodies were absent, and kidney biopsy specimens showed severe interstitial infiltrates but few tubular lesions.

Phenindione-induced AIN is generally associated with the development of hepatitis, which can be fatal.

Allopurinol-induced AIN appears to occur more often in patients with CKD and is usually seen in association with rash and liver dysfunction and sometimes with full manifestations of Stevens-Johnson syndrome.¹³ This severe allergic reaction is primarily observed in patients with the human leukocyte antigen B58 genotype.

AIN caused by NSAIDs is associated with nephrotic syndrome in about 75% of cases. Patients are usually older than 50 years, and although it has been observed with all NSAIDs, including cyclooxygenase 2-selective inhibitors, half of the cases have been reported with fenopfen. Most occurrences develop after the patient has taken NSAIDs for months (6 months on average), but AIN can occur within days or after more than a year. With the exception of the heavy proteinuria and associated edema, the presentation of these patients is similar to that of patients with other forms of drug-induced AIN (Fig. 64.4), although extrarenal symptoms are present in only about 10% of patients. Kidney disease caused by NSAIDs must be differentiated from other NSAID kidney manifestations, including hemodynamically mediated AKI, papillary necrosis, and NSAID-induced membranous nephropathy. Drugs other than NSAIDs can rarely induce AIN associated with the nephrotic syndrome; a few cases have been reported after administration of ampicillin, rifampin, lithium, interferon, phenytoin, pamidronate, and D-penicillamine.

Clinical Presentation of AIN and Nephrotic Syndrome Associated With NSAID Use

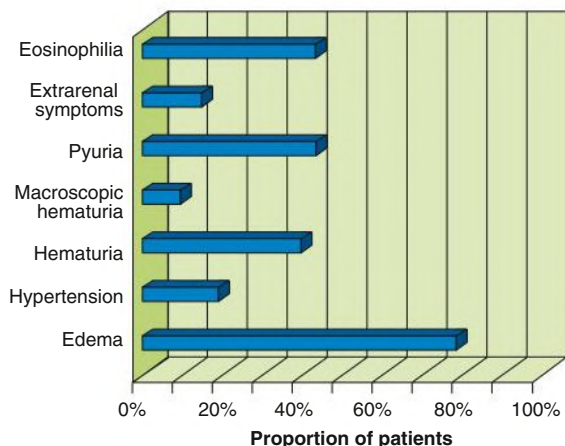


Fig. 64.4 Clinical Presentation of Acute Interstitial Nephritis (AIN) and Nephrotic Syndrome Associated With Nonsteroidal Antiinflammatory Drug (NSAID) Use. Data were pooled from different case reports including more than 60 patients.

PPIs, such as omeprazole or pantoprazole, are commonly recognized as a cause of AIN. Symptoms are nonspecific, although approximately 10% of subjects present with fever, chills, and anorexia. Consistent with the fact that drug-induced AIN can lead to permanent kidney damage, large observational studies have shown that the use of PPIs is associated with an increased risk for development or progression of CKD, and development of AIN is viewed as being the likely underlying mechanism.¹⁴

Target immune therapies have transformed the treatment and outcomes of many cancers in the last few years but not without multiple immune-mediated adverse side effects, including AKI.¹⁵ In one large study, immune checkpoint inhibitor-associated AKI usually occurred at an average of 14 weeks (range, 6–37 weeks), and the majority had other drugs known to cause AIN (69%) and often with multiple checkpoint inhibitors used on patients at the same time, with the majority recovering (85%). A small biopsy series demonstrated that 92% of patients had tubulointerstitial nephritis as the cause of the AKI, which was associated with persistent mild kidney impairment at 12 months.¹¹ The use of kidney biopsy versus just an empiric steroid pulse when any AKI occurs associated with checkpoint inhibitor use is controversial, but biopsy may aid decision making about whether to continue giving a checkpoint inhibitor and/or prescribe corticosteroids.¹⁶

Pathology

The hallmark of AIN is the presence of inflammatory infiltrates in the kidney interstitium (Fig. 64.5). These infiltrative lesions are often patchy, predominating in the deep cortex and outer medulla, but they can be diffuse in severe cases. They are composed mostly of T cells and monocytes-macrophages, but plasma cells, eosinophils, and a few neutrophilic granulocytes also may be present. The relative number of CD4⁺ T cells and CD8⁺ T cells is variable from one patient to another. In some cases, T lymphocytes infiltrate across the TBM and between tubular cells, mainly in distal tubules, and the resulting lesion is referred to as *tubulitis*.

In some cases of drug-induced AIN, kidney biopsy shows interstitial granulomas (Fig. 64.6). These granulomas are usually sparse and nonnecrotic, with few giant cells, and are associated with

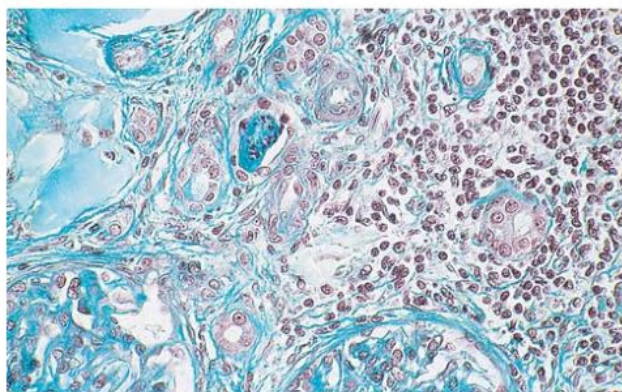


Fig. 64.5 Drug-Induced Acute Interstitial Nephritis. On light microscopy, the characteristic feature is interstitial infiltration with mononuclear cells, with normal glomeruli. It is usually associated with interstitial edema and tubular lesions. (Courtesy Dr. B. Mougenot, Paris VI University, Paris.)

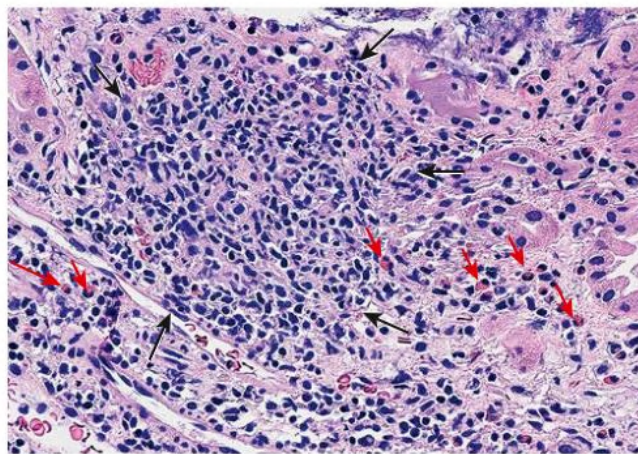


Fig. 64.6 Drug-induced granulomatous acute interstitial nephritis (AIN). The patient was taking omeprazole over the counter and presented with fever, chills, malaise, and cough. Workup showed a serum creatinine of 4.5 mg/dL (400 μ mol/L) and 15.5-cm enlarged kidneys that were pushing up on the diaphragm, causing the cough. Biopsy shows AIN with eosinophils (red arrows), lymphocytes, and early granuloma formation (demarcated by black arrows). (Courtesy R.J. Johnson, University of Colorado, Denver.)

nongranulomatous interstitial infiltrates. Granulomas are also found in AIN related to infection (Box 64.2), sarcoidosis, Sjögren syndrome, and granulomatous polyangiitis.

Interstitial infiltrates are always associated with interstitial edema, which is responsible for separating the tubules (see Fig. 64.5). They can also be associated with focal tubular lesions, which range from mild cellular alterations to extensive necrosis of epithelial cells, and are sometimes associated with a disruption of the TBM. These tubular lesions usually predominate where the inflammatory infiltrates are most extensive.

Tubulointerstitial lesions are not associated with vascular or glomerular lesions. Even in AIN associated with a nephrotic syndrome, glomeruli appear normal on light microscopy; glomerular lesions are similar to those seen in minimal change disease (see Chapter 18).

In most patients with AIN, kidney biopsy specimens do not show immune deposits, and both IF and electron microscopy are negative. Nevertheless, staining of the tubular or capsular basement membrane

BOX 64.2 Infections That Can Be Associated With Acute Interstitial Nephritis

Bacteria	Hantavirus
<i>Brucella</i> spp.	Hepatitis A virus
<i>Campylobacter jejuni</i>	Hepatitis B virus
<i>Corynebacterium diphtheriae</i>	Herpes simplex virus
<i>Escherichia coli</i>	Human immunodeficiency virus
<i>Legionella</i> spp.	Measles virus
<i>Leptospira</i> spp.	Polyomavirus
<i>Mycobacterium tuberculosis</i> ^a	Rickettsia
<i>Salmonella</i> spp. ^a	
<i>Staphylococcus</i> spp.	Parasites
<i>Streptococcus</i> spp.	<i>Toxoplasma</i> spp. ^a
<i>Yersinia pseudotuberculosis</i>	<i>Leishmania donovani</i>
Viruses	Other
Adenovirus	<i>Chlamydia</i> spp.
Cytomegalovirus	<i>Mycoplasma</i> spp.
Epstein-Barr virus ^a	

^aInfections that can induce granulomatous acute interstitial nephritis.

for IgG or complement may occasionally be seen by IF; the staining pattern is either granular or linear (Fig. 64.7). Linear fixation of IgG along the TBM indicates the presence of antibodies directed against membrane antigens or against drug metabolites bound to the TBM, and circulating anti-TBM antibodies have been detected in some cases. These linear deposits are seen mostly in patients taking NSAIDs, phenytoin, or allopurinol.

Diagnosis

The most accurate way to diagnose AIN is by kidney biopsy. However, both eosinophiluria and gallium scanning have been suggested as helpful in making the diagnosis.

Eosinophils can be detected in urine and, although eosinophiluria is frequently used to corroborate the diagnosis of drug-induced AIN, review of published series shows that this test has rather poor sensitivity and a low positive predictive value, even when only patients with AKI are considered. In the largest and most recent series, only 36% (26 of 73) of patients with drug-induced AIN had eosinophiluria, and 28% (20 of 69) of patients with acute tubular necrosis also had eosinophiluria (Fig. 64.8).¹⁷ Eosinophiluria may also occur in patients with diseases such as proliferative or crescentic glomerulonephritis (GN), atheroembolic kidney disease, urinary tract infection, urinary schistosomiasis, and even prerenal AKI.¹⁸ Thus, eosinophiluria should be abandoned as a test for AIN. Firm diagnosis requires kidney biopsy and/or a clinical response on withdrawal of the drug of concern.

An increased kidney uptake of gallium-67 (⁶⁷Ga) has been reported in AIN.¹⁹ Analysis of available series shows that in 53 patients with AIN, 83% had an abnormal kidney scan (maximum after 48 hours), whereas it was normal in 17 of 18 patients with acute tubular necrosis. However, these studies were small and retrospective, and ⁶⁷Ga kidney scanning is not specific for AIN and may be abnormal in patients with pyelonephritis, cancer, or glomerular diseases. We do not recommend use of gallium scanning as a diagnostic tool.

Because the clinical presentation of AIN is variable and because noninvasive diagnostic procedures have limitations, kidney biopsy is often essential for the diagnosis of AIN. Several studies have shown that prebiopsy diagnosis may be incorrect in a substantial number of patients.

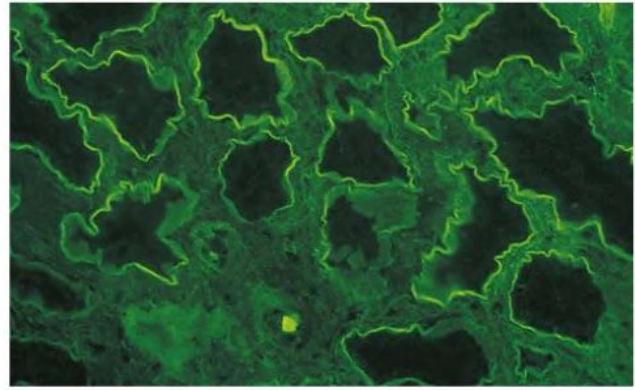


Fig. 64.7 Linear Deposits of Immunoglobulin G in Drug-Induced Acute Interstitial Nephritis. Deposits along the tubular basement membrane (TBM) are shown on immunofluorescence microscopy. These antibodies recognize either a component of the TBM or a drug metabolite bound to the TBM. (Courtesy Dr. B. Mougenot, Paris VI University, Paris.)

Identification of the Causative Drug

Identification of the causative drug is relatively easy when AIN occurs in a patient taking only one drug. However, patients are often taking more than one drug capable of inducing AIN and in a recent study,⁶ 30% of cases had no offending drug clearly implicated. Two biologic tests have been used, primarily in research laboratories, to help identify the causative drug: the lymphocyte stimulation test and the identification of circulating antidrug antibodies. However, neither test appears to be clinically useful at this time because of issues with sensitivity and specificity.

Natural History

Drug-induced AIN was long considered benign, with complete recovery of kidney function if the inciting agent was removed. However, recent studies show that the course of AIN is not always benign, and serum creatinine does not return to baseline in about 40% to 50% of patients.^{4,20} Even in those with kidney recovery, an increase in serum creatinine can persist for several weeks. Unfortunately, there are few known prognostic predictors. The severity of kidney failure does not appear to be linked with the prognosis.⁴ It has been suggested that the presence on kidney biopsy of diffuse neutrophil- or macrophage-rich infiltrates, interstitial granulomas, or tubular atrophy is associated with a poor prognosis, but this has not been consistently found in all series. The best prognostic predictors may actually be the duration of AKI and the severity of interstitial fibrosis.

Treatment

In addition to removal of the inciting agent, which is essential and should be done as soon as possible, corticosteroids have been used to treat AIN. Most commonly, patients received high-dose oral corticosteroids sometimes associated with pulses of intravenous methylprednisolone. In the recent decade, properly controlled trials still suggest that corticosteroids improve outcomes, particularly in idiopathic AIN.²⁰ The largest recent study⁶ confirmed short-course therapy (3 weeks) was as effective as a prolonged course (8 weeks).

In different series, corticosteroids rapidly induced a reduction in serum creatinine in patients whose kidney function did not improve within about 1 week after discontinuation of the inciting agent. It is interesting that in patients with NSAID-induced AIN, corticosteroids do not seem to modify the course of the nephrotic syndrome.

Eosinophiluria in the Diagnosis of Drug-Induced Acute Interstitial Nephritis

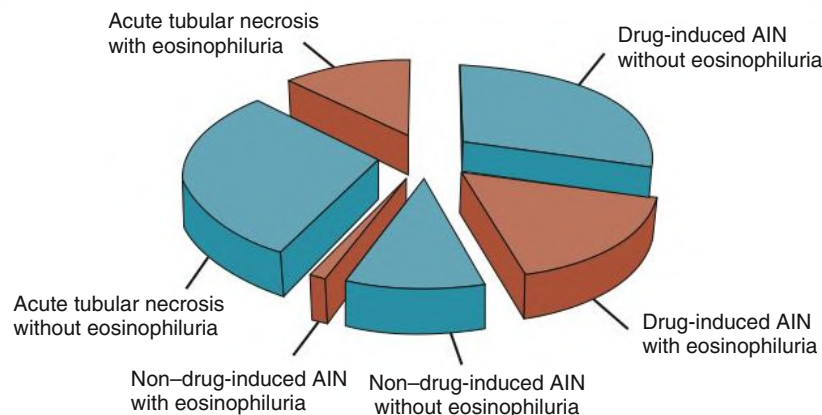


Fig. 64.8 Eosinophiluria in the Diagnosis of Drug-Induced Acute Interstitial Nephritis (AIN). Data from a recent and large retrospective series confirm that eosinophiluria (defined by the presence of 1% or more eosinophils in urine) lacks sensitivity and specificity for the diagnosis of drug-induced AIN in patients with acute kidney injury. (From Lazarus B, Chen Y, Wilson FP, et al. Proton pump inhibitor use and the risk of chronic kidney disease. *JAMA Intern Med.* 2016;176[2]:238–246.)

On the basis of anecdotal case reports, some authors have advocated use of mycophenolate mofetil in patients resistant to corticosteroids.²¹

We recommend administration of a 3-week course of prednisolone in patients whose kidney function fails to improve within 1 week after discontinuation of the inciting drug and rapidly return to baseline values, provided the diagnosis of AIN has been confirmed by kidney biopsy.⁶ We initiate treatment with a dose of prednisone of 1 mg/kg/day and not higher than 60 mg/day, and after 1 to 2 weeks, we progressively taper the dose so that the total duration of treatment is 4 to 6 weeks.

ACUTE INTERSTITIAL NEPHRITIS SECONDARY TO INFECTIOUS DISEASES

Infections were once the most common cause of AIN, but the frequency of AIN induced by an infection has decreased with the widespread use of antibiotics. Nevertheless, the diagnosis of infectious AIN should not be overlooked, and AIN occurring in patients treated with antibiotics should not always be attributed to the drug.

Infectious agents can cause kidney parenchymal inflammation by direct infection, resulting in acute pyelonephritis (see [Chapter 53](#)). However, many infectious agents may induce an immunologically mediated AIN in the absence of direct invasion (see [Box 64.2](#)). In this case the clinical presentation depends mostly on the underlying infection. On histologic examination, lesions are identical to those described for drug-induced AIN, and they also can occasionally result in granulomas (see [Box 64.2](#)). Infection-associated AIN usually resolves with the treatment of the underlying infection, and corticosteroid therapy is not recommended.

An important cause of infection-associated AIN is hantavirus.²² Hantavirus infections occur worldwide and are responsible for a disease that has been known as *hemorrhagic fever with renal syndrome*, *epidemic hemorrhagic fever*, and *nephropathia epidemica*. Rodents are the main reservoir of the virus, and humans are probably infected by the airborne route. Besides fever, fatigue, and muscle aches (sever lumbar pain that is sometime interpreted as nephrolithiasis), which are

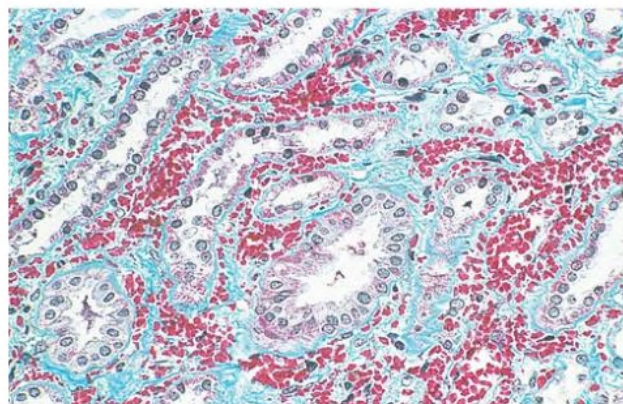


Fig. 64.9 Acute Interstitial Nephritis Secondary to Hantavirus Infection. Vascular congestion and foci of medullary hemorrhage are suggestive of the diagnosis. (Courtesy Dr. B. Mougnot, Paris VI University, Paris.)

observed in all patients, extrarenal symptoms often include headache, lightheadedness, abdominal pain, nausea and vomiting, and thrombocytopenia; the last can be responsible for hemorrhagic complications. AKI is almost always associated with proteinuria, sometimes in the nephrotic range, and with hematuria. Kidney biopsy is usually not required for the diagnosis but if performed reveals not only interstitial inflammatory infiltrates, which predominate in the medulla, but also vascular congestion and interstitial bleeding ([Fig. 64.9](#)). In 50% of patients, IF studies show granular immune deposits along the TBM and within glomeruli. Kidney function usually improves after a few days, and complete recovery is the rule. Nevertheless, in severe cases, recovery can be complicated by the occurrence of hemorrhagic complications or severe shock. The diagnosis is based on serologic test results, which become positive early (within weeks) in the course of the disease.

Tubulointerstitial lesions are common in human immunodeficiency virus (HIV)-positive patients who undergo a kidney biopsy for

AKI. Interstitial infiltrates are often associated with glomerular lesions, but they can be isolated. These forms of AIN have been observed in both White and Black patients, and they might be related not only to drugs and opportunistic infections but also to the HIV infection itself.^{23,24}

ACUTE INTERSTITIAL NEPHRITIS ASSOCIATED WITH SYSTEMIC DISEASES

Sarcoidosis

In sarcoidosis, reductions in the glomerular filtration rate (GFR) usually occur as a complication of hypercalciuria and hypercalcemia, but granulomatous AIN associated with sarcoidosis also may occur (Fig. 64.10).^{25,26} The presentation is usually that of AKI, which can be isolated or associated with mild proteinuria and sterile leukocyturia. It is associated with extrarenal symptoms of sarcoidosis in about 90% of patients, most frequently with lymphadenopathy and lung, eye, or liver involvement. Nevertheless, only slightly more than half of the patients have hilar lymphadenopathy or pulmonary interstitial fibrosis at the time of diagnosis.²⁶ Treatment with high-dose corticosteroids quickly improves kidney function, but most patients do not recover completely. The starting dose should be prednisone 1 mg/kg/day and not higher than 60 mg/day, and corticosteroid therapy should be tapered slowly and not withdrawn before at least 1 year to prevent relapses. Whereas some authors advocate long-term maintenance therapy with low-dose corticosteroids, we usually stop corticosteroids after 2 to 3 years. Because of the risk for late relapse, these patients should be observed for a prolonged time.

Sjögren Syndrome

Clinically significant interstitial nephritis is rare in Sjögren syndrome and usually results in chronic tubular dysfunction.²⁷ Some patients may present with severe symptomatic hypokalemia with distal renal tubular acidosis. Rarely, Sjögren syndrome presents with AKI because of AIN, which is often responsive to treatment with high-dose corticosteroids.

Systemic Lupus Erythematosus

About two-thirds of kidney biopsy samples in patients with systemic lupus show tubulointerstitial involvement, but significant tubulointerstitial injury in the setting of minimal glomerular abnormalities is rare, with only about 10 cases reported in the literature.²⁸ In these cases, kidney biopsy shows typical features of AIN on light microscopy and IF staining always discloses immune deposits along the TBM, usually with a granular pattern. Kidney function improves after high-dose corticosteroids, and additional immunosuppressive drugs are not usually required. Azathioprine also has been used as a corticosteroid-sparing agent.

Immunoglobulin G4-Related Disease

IgG4-related disease is a newly recognized systemic autoimmune disease that predominantly affects men older than 50 years (see Chapter 65). In the kidney, it is most commonly responsible for AIN characterized by the presence of interstitial infiltrates rich in IgG4-positive plasma cells and immune deposits along the TBM.^{29,30} This interstitial nephritis can be associated with membranous nephropathy, with kidney inflammatory masses visible on imaging, or with ureteral obstruction. It usually responds rapidly to treatment with corticosteroids.

Other Systemic Diseases

Among patients with cryoglobulinemia and AKI, a few exhibit significant interstitial inflammatory infiltrates associated with granular immune deposits in the interstitium and along the TBM. This

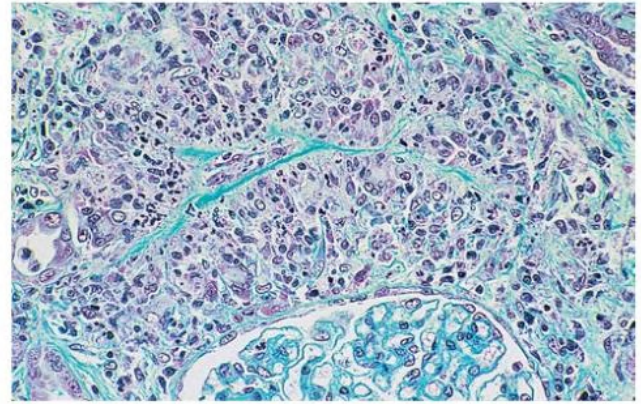


Fig. 64.10 Granulomatous acute interstitial nephritis in a patient with sarcoidosis. (Courtesy Dr. B. Mougenot, Paris VI University, Paris.)

AIN is usually associated with characteristic glomerular lesions and, rarely, lesions of the arterioles, and treatment is the same as for cryoglobulinemia-induced GN (see Chapter 26).

Most kidney lesions associated with small-vessel vasculitis consist of both extracapillary GN and tubulointerstitial nephritis (see Chapter 26). Nevertheless, a few patients with AIN and minimal glomerular lesions have been described.

ACUTE INTERSTITIAL NEPHRITIS ASSOCIATED WITH MALIGNANCY

Infiltration of kidney parenchyma by malignant cells is common in patients with leukemia or lymphoma. Most of the time, this infiltration is clinically silent (or causes enlarged kidneys), but a few patients with AKI have been described.³¹ Chemotherapy or radiotherapy may rapidly improve kidney function in these patients, but before these treatments are started, it is important to exclude more common causes of AKI associated with cancer (see Chapter 67).

IDIOPATHIC ACUTE INTERSTITIAL NEPHRITIS

More than 50 cases of idiopathic AIN with anterior uveitis have been reported (tubulointerstitial nephritis and uveitis [TINU] syndrome).³² This syndrome is found most commonly in females of pubertal age but also can occur in pubertal males and in adults. Initial symptoms may be ocular, with ocular pain and visual impairment, or pseudo-viral, with fever, myalgia, and asthenia. AIN is responsible for AKI, ranging from mild to severe, that may be associated with abnormal urinalysis. Kidney biopsy shows diffuse interstitial inflammatory infiltrates, almost always without granulomas and without immune deposits. In children, kidney prognosis is excellent, and serum creatinine usually returns to baseline values within a few weeks, with or without corticosteroid therapy. In adults the kidney prognosis seems to be less favorable, and corticosteroid therapy might be useful in preventing evolution to chronic kidney impairment. Uveitis, which can occur at any time in respect to AIN, is usually responsive to topical corticosteroids, but it may relapse without any recurrence of AIN.

Cases of idiopathic AIN without uveitis have been reported. IF studies of kidney biopsy specimens can show linear deposits of IgG along the TBM, granular deposits of IgG along the TBM, or no immune deposits, suggesting this entity is heterogeneous. The treatment of patients with idiopathic AIN is still controversial. Patients who receive corticosteroids usually show a dramatic improvement of kidney function, but others have recovered normal kidney function without any treatment.

ACUTE INTERSTITIAL NEPHRITIS IN KIDNEY TRANSPLANTS

Acute rejection is by far the most common cause of AIN in kidney allograft recipients (see [Chapter 109](#)). Nevertheless, AIN can be induced by drugs or infections. Cases of drug-induced AIN have been

reported even in the first weeks after transplantation, when immunosuppression is maximal.³³ Among patients with infectious AIN, the frequency of polyomavirus-induced AIN appears to be increasing, and it should be suspected in patients with acute deterioration of kidney function and so-called decoy cells in urine (see [Chapter 110](#)).³⁴

SELF-ASSESSMENT QUESTIONS

1. A 65-year-old man presents with unexplained AKI. A kidney biopsy shows the presence of diffuse inflammatory infiltrates within the interstitium, leading to the conclusion that the AKI is a result of AIN. He has a recent history of treatment with an NSAID and a PPI for osteoarthritis. Which one of the following statements is *correct*?
 - A. Both NSAIDs and PPIs are among the classes of drugs most commonly responsible for AIN.
 - B. Only very few cases of AIN have been attributed to treatment with a PPI, and the AIN was most likely induced by treatment with an NSAID.
 - C. Only very few cases of AIN have been attributed to treatment with an NSAID, and the AIN was most likely induced by treatment with a PPI.
 - D. Only very few cases of AIN have been attributed to treatment with a PPI or an NSAID, and the AIN is unlikely to have been induced by treatment with an NSAID or a PPI.
2. A 75-year-old woman presents with AKI associated with mild proteinuria (albumin-to-creatinine ratio 500 mg/g), microscopic hematuria, and pyuria. AKI developed while she was being treated with a fluoroquinolone, and her treating physician suspects that it was caused by AIN induced by the fluoroquinolone. Which one of the following statements is *correct*?
 - A. The patient should be tested for eosinophiluria. If less than 1% of urinary white blood cells are eosinophils, the diagnosis of AIN can almost certainly be ruled out.
 - B. The patient should undergo a kidney gallium scan. If there is no kidney uptake, the diagnosis of AIN can almost certainly be ruled out.
 - C. The presence of an elevated ratio of albumin to creatinine is sufficient to rule out the diagnosis of drug-induced AIN in a patient who previously did not have albuminuria and who has not been treated with an NSAID.
 - D. A kidney biopsy is the only test that can be performed to reliably rule out or confirm the diagnosis of AIN.
3. A 30-year-old woman develops AKI while being treated with penicillin. A kidney biopsy shows diffuse interstitial infiltrates, confirming the diagnosis of AIN. Her serum creatinine has been stable at 3.0 mg/dL for the last 5 days. Which one of the following statements is *correct*?
 - A. The prognosis for drug-induced AIN is excellent, and full recovery should be expected within 6 weeks after removal of the inciting agent.
 - B. The prognosis for drug-induced AIN is excellent in patients who do not have granulomas on renal biopsy, and full recovery should be expected.
 - C. The prognosis for drug-induced AIN is excellent in patients who do not require dialysis, and full recovery should be expected if serum creatinine does not continue to rise.
 - D. The course of drug-induced AIN is not always benign, and it leads to chronic kidney disease in at least 40% of cases.

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